Design of a Low Cost Pitch Canard-Controlled Active Stability System

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Abstract— **Active trajectory control systems are widely used to improve sounding rocket performance at higher apogees, where small perturbations can produce significant dispersion in both altitude and landing location. Although this approach has been explored by student teams at universities such as Leeds, Santa Clara University, and TU Delft, competition records from ENMICE indicate that participating Mexican teams have so far relied exclusively on passive stabilization. This article presents the design and modeling of a low-cost active pitch control prototype for a high-power rocket powered by a Class J solid-propellant motor developed by the author. A canard-based configuration was selected over thrust vector control due to the limited internal volume of the 3-inch airframe and the actuation requirements imposed by the motor's short burn time. Following a systems engineering approach based on a V-model methodology, the vehicle's mass properties and aerodynamic derivatives were characterized using CAD-based inertia estimation and CFD simulation, and used to derive a linearized pitch plant model. A PID controller was synthesized via triple-pole placement on the linearized model and refined through direct numerical optimization using an integrated-squared-error (ISE) cost function with an overshoot penalty term. Closed-loop simulation results show that control authority scales with dynamic pressure, resulting in longer correction times as coast velocity decreases, and that the maximum canard deflection tested exhibits a non-monotonic aerodynamic response at high combined incidence angles, whose physical origin (flow separation versus numerical uncertainty) remains to be confirmed. These findings define a set of concrete improvements, including actuator dynamics modeling, deflection scheduling, and gain scheduling with dynamic pressure, that are proposed prior to hardware-in-the-loop and flight testing.**

I. INTRODUCTION

Trajectory active control systems are widely used for improving sounding rocket trajectories as they reach higher apogees, where a small perturbation can create a significant difference both in altitude reached and horizontal dispersion. According to ENMICE [1], Mexican sounding rocket trajectories are not controlled actively but passively so far, which creates an area of improvement in active trajectory control systems in high-power rocketry. However, this topic has been explored by student teams at universities such as Leeds [2], Santa Clara University [3] and TU Delft [4].

In this regard, the objective of this article is to develop an initial prototype for active trajectory control of a high-power rocket powered by a Class J solid-propellant motor developed by the author. The prototype uses low-cost, readily available materials and is designed following a systems engineering approach.

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Although thrust vector control (TVC) is an alternative for trajectory correction, it was discarded for this vehicle: the limited internal volume available in a 3-inch diameter airframe constrains the size of a gimbaled nozzle assembly and its actuation mechanism, and the motor's short burn time would likely require higher-torque, faster-response servomotors than those available within the project's budget. Canard-based control was therefore selected as the more practical alternative given these constraints.

The vehicle configuration further defines the design space for this control system. The fuselage is made of fiberglass, and the control surfaces (canards) are 3D-printed using a combination of carbon fiber and PETG filament, chosen to balance structural stiffness against the actuation torque available from low-cost servomotors.

With the goal of reducing horizontal dispersion along the trajectory and supporting predictable vehicle recovery, a four-surface configuration was selected. The scope of this work is limited to the study of pitch control using two of these surfaces.

The remainder of this paper is organized as follows: Section II presents the system requirements, architecture and methodology. Section III describes the system design: vehicle's mass properties, aerodynamic characterization, pitch plant model and the PID control design, and closed-loop simulation results. Section IV covers the hardware implementation, and Section V discusses testing results and limitations. Section VI concludes the paper and outlines future work.

II. SYSTEM REQUIREMENTS, ARCHITECTURE AND METHODOLOGY

A. System Requirements

Accordingly with previously established goals, a requirement definition was conducted to translate high-level objectives into technical requirements.

- Active control enabled only during coasting phase.
- Ensure modularity in avionics bay.
- Maintain structural integrity of the complete assembly within the specified Mach 0.5 design regime.
- Control pitch perturbations due to crosswind.
- Attitude determination through sensor fusion.
- Low cost hardware.

B. Architecture

According to Fortescue [5], this system has been divided in Structural, Attitude Control, Electrical Power, Data Handling and Processing Systems.

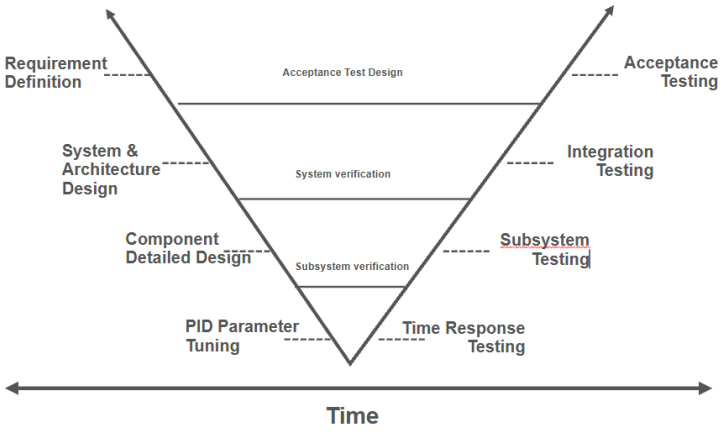


Fig. 1. V Model Diagram for the System.

C. Methodology

Once the requirements and architecture were established, a methodology was proposed based on the project life cycle approach outlined in NASA's Systems Engineering Handbook; however, in addition to the suggested phases, the following flowchart shows the steps taken for the design and development of this system.

III. System Design

A. Vehicle Mass & Size Properties

The vehicle mass properties were obtained from the fully assembled CAD model in SolidWorks. The mass distribution was evaluated with respect to the vehicle body axes and the corresponding center of gravity. The total vehicle mass considered for the control-system analysis is

$$m = 3.5 \text{ kg}. \tag{1}$$

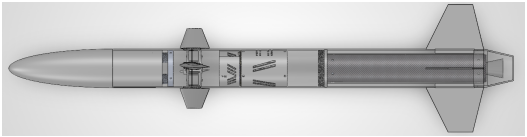


Fig. 3. Rocket CAD

The pitch-axis moment of inertia obtained from the CAD model is

$$I_{yy} = 0.1017 \text{ kg m}^2. \tag{2}$$

This value is used throughout the pitch-dynamics model to relate the aerodynamic pitching moment to the angular acceleration of the vehicle.

The vehicle has an overall length of approximately

$$L = 1.00 \text{ m}, \tag{3}$$

and a nominal fuselage diameter of

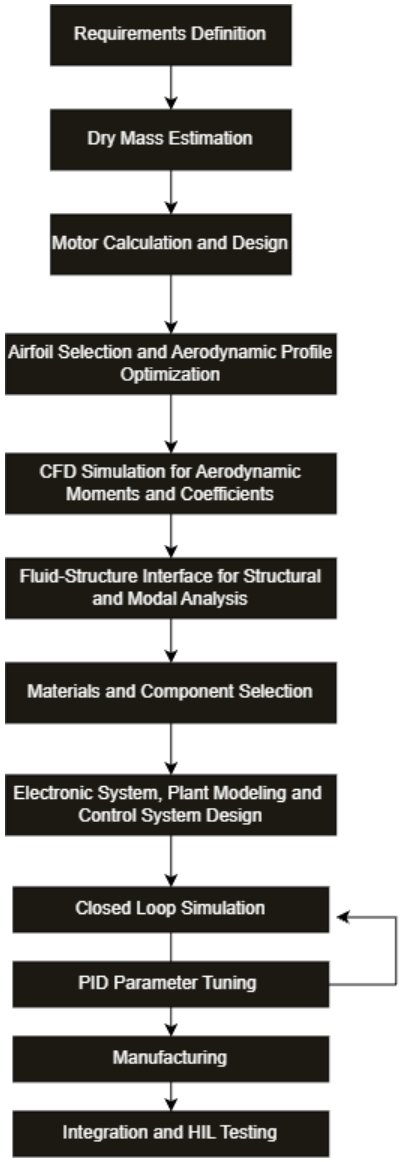


Fig. 2. Implemented Design Methodology.

$$D = 0.078 \text{ m}. \tag{4}$$

The center of gravity is located at approximately

$$x_{CG} = 0.605 \text{ m} \tag{5}$$

measured from the nose of the vehicle. The center-of-pressure location obtained from the aerodynamic characterization is used together with the center of gravity to evaluate the longitudinal static stability of the vehicle.

The control canards are located at approximately

$$x_c = 0.345 \text{ m} \tag{6}$$

from the nose. Therefore, the longitudinal distance between the canard location and the center of gravity is

$$l_c = x_{CG} - x_c = 0.605 - 0.345 = 0.260 \text{ m}. \tag{7}$$

This distance provides the longitudinal moment arm through which the aerodynamic forces generated by the control surfaces contribute to the vehicle pitching moment.

B. Canard Geometry and Airfoil Selection

The vehicle incorporates four aerodynamic surfaces distributed around the fuselage. Two of these surfaces are considered as the active control surfaces for the pitch axis, while the remaining surfaces contribute primarily to the overall aerodynamic configuration and stability.

The control canards are positioned approximately 345 mm from the nose of the vehicle. Their three-dimensional geometry was developed using a symmetric NACA 0012 airfoil.

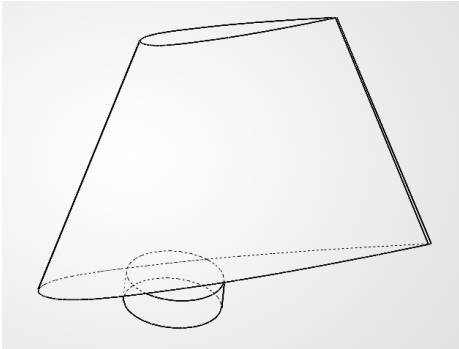


Fig. 4. Canard CAD

The NACA 0012 belongs to the NACA four-digit airfoil family. Its designation indicates a maximum camber of zero percent of the chord and a maximum thickness of twelve percent of the chord. Consequently, the mean camber line coincides with the chord line, resulting in a geometrically symmetric airfoil.

The maximum thickness is therefore

$$t_{max} = 0.12c, \tag{8}$$

where c denotes the local chord.

The symmetric geometry was selected because the control surface must operate in both positive and negative deflection directions. Unlike a cambered airfoil, the NACA 0012 provides the same geometric configuration for both directions of operation and establishes a well-defined neutral configuration.

At zero angle of attack and zero control-surface deflection, the symmetric profile does not introduce a lift contribution associated with airfoil camber. Consequently, the aerodynamic response associated with the control surface is primarily determined by the angle of attack and the imposed deflection.

This characteristic is particularly convenient for a control surface because the same geometry can generate aerodynamic effects in opposite directions when its deflection is reversed. The use of a symmetric airfoil also simplifies the manufacturing process because the upper and lower surface geometries are mirror images about the chord line.

The canard geometry has a root chord of approximately

$$c_r = 60 \text{ mm}, \tag{9}$$

and an outer chord of approximately

$$c_t = 30 \text{ mm}, \tag{10}$$

with an approximate span of

$$b = 40 \text{ mm}. \tag{11}$$

The resulting three-dimensional CAD model has a surface area of

$$A_{CAD} = 4417.74 \text{ mm}^2. \tag{12}$$

The reported CAD area corresponds to the three-dimensional surface area of the modeled canard and should therefore be distinguished from the projected planform area used as an aerodynamic reference quantity.

The canards are geometrically arranged in opposite pairs around the fuselage. This symmetric arrangement minimizes aerodynamic asymmetries when the control surfaces are in their neutral position and provides a consistent geometric basis for the pitch-control configuration.

The center of pressure was obtained from the aerodynamic model at $x_{CP} = 0.71m$ and its longitudinal position was compared with the center of gravity at $x_{CG} = 0.605$ m. This relative position was subsequently used to evaluate the longitudinal static stability of the vehicle.

$$SM = \frac{x_{CP} - x_{CG}}{D} = 0.13 \tag{13}$$

C. Aerodynamic Characterization

The aerodynamic characteristics of the vehicle were obtained using Computational Fluid Dynamics (CFD) simulations performed in ANSYS Fluent. The aerodynamic analysis considered variations in angle of attack, α, and canard deflection angle, δ.

The reference area used for the aerodynamic coefficients is the frontal cross-sectional area of the fuselage,

$$A_{ref} = \frac{\pi D^2}{4}, \tag{14}$$

which, for a diameter of $D = 0.078$ m, gives

$$A_{ref} \approx 4.78 \times 10^{-3} \text{ m}^2. \tag{15}$$

The aerodynamic pitching moment is represented in nondimensional form by the pitching moment coefficient C_m. The dimensional pitching moment is related to the coefficient through

$$M_y = qA_{ref}d_{ref}C_m, \tag{16}$$

where q is the dynamic pressure, d_{ref} is the reference length, and C_m is the pitching moment coefficient. For this CFD analysis a cylindric control volume was used with 10L upstream, 15L downstream, and a radius of 5D, where L

is the nominal length of the rocket and D is the nominal diameter. A refined mesh was used in areas of interest such as the leading edge of the fins for passive control, the total surface area of the canards for active control, and the nose of the rocket, with the respective node and element limitations of the student version of ANSYS. A study of mesh independence was also conducted, but it will not be included in this paperwork.

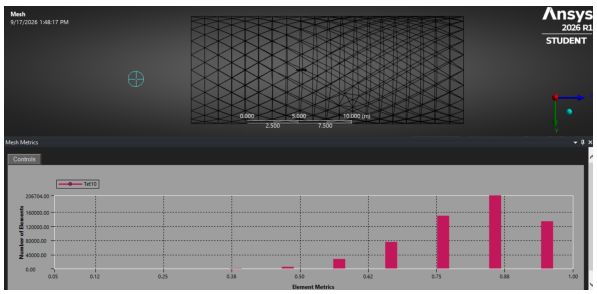


Fig. 5. Control Volume Mesh

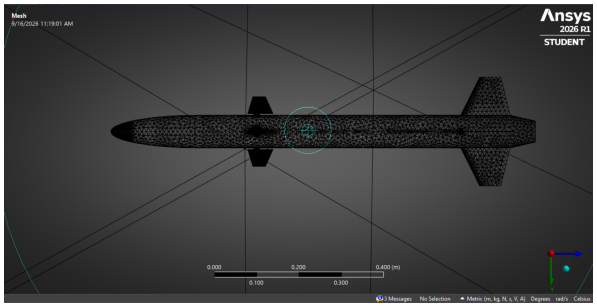


Fig. 6. Rocket Mesh

The dynamic pressure is defined as

$$q = \frac{1}{2}\rho V^2, \tag{17}$$

where ρ is the local air density and V is the vehicle velocity.

For example, using standard sea-level air density $\rho = 1.225\ \mathrm{kg/m^3}$ and a representative velocity of $V = 170\ \mathrm{m/s}$,

$$q = \frac{1}{2}(1.225)(170)^2 \approx 17.7\ \mathrm{kPa}. \tag{18}$$

We can verify this using the dynamic pressure contour obtained with ANSYS Fluent. In this contour, we can see that for the active control surfaces that do not exhibit deflection, the dynamic pressure is in the range of 17 kPa.

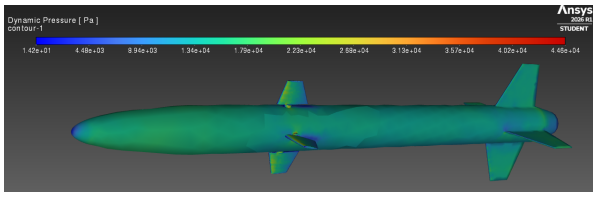


Fig. 7. Dynamic Pressure Contour

α	δ	C_p	C_D	M_x	C_{mx}	M_y	C_{my}
-5º	0	0.68	0.4	-0.77	-0.009	-10.18	-0.12
-2º	0	0.667	0.42	0.39	0.004	-3.26	-0.03
0	0	0.71	0.428	0.613	0.007	-0.114	-0.001
2º	0	0.658	0.421	0.76	0.009	2.7	0.032
5º	0	0.67	0.4	0.55	0.006	9.524	0.113
-5º	5	0.71	0.56	-0.12	-0.001	15.3	0.18
-2	5	0.52	0.468	-0.41	-0.004	0.05	0.0524
0	5	0.92	0.45	0.12	0.001	7.43	0.094
2	5	0.82	0.467	0.41	0.004	10.11	0.119
5º	5	0.61	0.59	1.39	0.0164	-0.63	-0.007
5º	10	0.57	0.42	-0.024	-0.002	22.64	0.26
2º	10	0.42	0.44	-0.52	-0.006	16.11	0.19
0	10	0.41	0.45	-0.97	-0.011	13.3	0.158
-2º	10	0.42	0.45	-0.25	-0.03	11.18	0.13
-5º	10	0.6	0.44	-0.74	-0.0089	5.48	0.064

TABLE I

AERODYNAMIC RESULTS FROM CFD ANALYSIS OF CANARD DEFLECTION (δ)

The CFD simulations were performed for multiple combinations of α and δ in order to characterize both the passive longitudinal response and the effect of canard deflection.

The resulting aerodynamic data for different canard deflection angles are presented in Table I.

The CFD results show a clear dependence of the vehicle pitching moment on both angle of attack and canard deflection. For $\delta = 0°$, the pitching moment changes sign as the angle of attack varies from negative to positive values. This behavior is consistent with the restoring aerodynamic response expected from a statically stable longitudinal configuration.

When the canards are deflected, a significant modification of the pitching moment is observed. At $\alpha = 0°$, for example, the simulated pitching moment changes from approximately -0.114 at $\delta = 0°$ to 7.43 at $\delta = 5°$ and 12.26 at $\delta = 10°$.

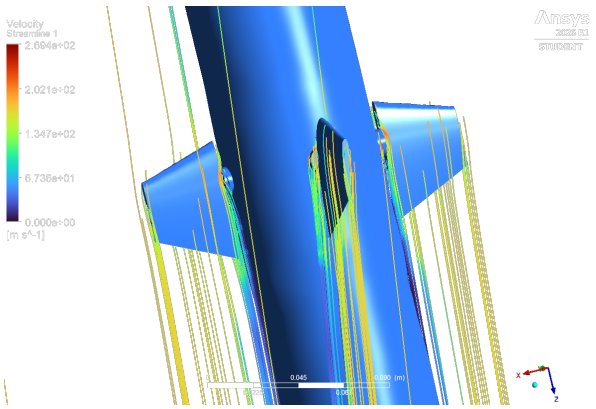


Fig. 8. Change in the fluid's trajectory due to canards deflection

These results demonstrate that the canards provide a significant aerodynamic contribution to the longitudinal moment of the vehicle. The relationship is not perfectly linear over the analyzed range, indicating that the aerodynamic response becomes nonlinear in certain cases.

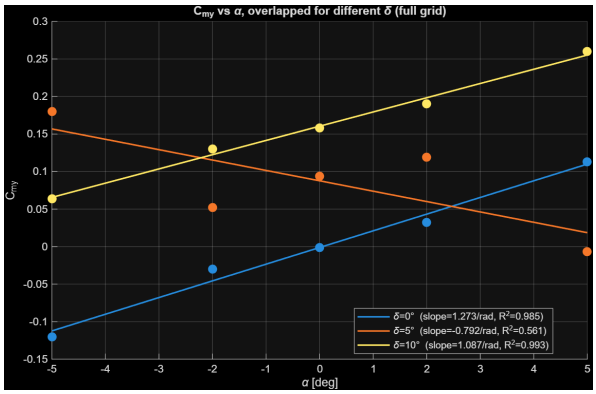


Fig. 9. C_{my} vs α

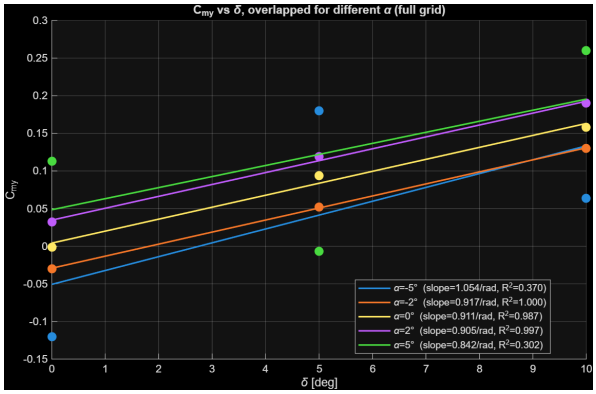


Fig. 10. C_{my} vs δ

D. Aerodynamic Moment Characterization

The pitch moment coefficient C_{my} was characterized through a complete CFD sweep grid over $\alpha \in \{-5, -2, 0, 2, 5\}$ and $\delta \in \{0, 5, 10\}$ (15 cases total). Linear regression was applied along both axes of the grid; results are summarized in Table II.

TABLE II
LINEAR REGRESSION RESULTS FOR C_{my} (COMPLETE 5×3 GRID)

Sweep	Slope [rad^{-1}]	R^2	n
C_{my} vs α ($\delta = 0$)	$+1.273$	0.985	5
C_{my} vs α ($\delta = 5$)	-0.792	0.561	5
C_{my} vs α ($\delta = 10$)	$+1.087$	0.993	5
C_{my} vs δ ($\alpha = -5$)	$+1.054$	0.370	3
C_{my} vs δ ($\alpha = -2$)	$+0.917$	1.000	3
C_{my} vs δ ($\alpha = 0$)	$+0.911$	0.987	3
C_{my} vs δ ($\alpha = 2$)	$+0.905$	0.997	3
C_{my} vs δ ($\alpha = 5$)	$+0.842$	0.302	3

This confirms that the aerodynamic nonlinearity is concentrated at the extremes of the tested angle-of-attack range and not in the small-perturbation region relevant to closed-loop pitch control during coasting flight. Accordingly, $C_{m\delta} \approx +0.91\,\text{rad}^{-1}$ (raw CFD sign) is adopted as the linearization point for the control-oriented model.

As previously discussed, the sign of C_{my} is inverted in the control-oriented model to be consistent with the vehicle's confirmed static stability margin, yielding $C_{m\delta} \approx -0.91\,\text{rad}^{-1}$ for use in Eq. (23).

As a result, the linearized model adopted in this work uses the $\alpha = 0$ sweep as the basis for $C_{m\delta}$, since it best represents the small-perturbation regime expected during closed-loop operation, and the α-dependence of $C_{m\delta}$ is documented as a modeling limitation (Subsection E).

E. Pitch Plant Model

Because the vehicle is aerodynamically stable by design, its passive response to a pitch perturbation is not that of a free rotating body, but rather that of a naturally restoring second-order system, analogous to a mass-spring-damper: the tail fins provide a restoring moment proportional to angle of attack, while the vehicle's own rotation is aerodynamically damped. The active control system does not need to synthesize stability from an unstable or neutral plant; instead, it corrects the vehicle's existing passive response to minimize trajectory dispersion.

The dimensional aerodynamic moment acting on the vehicle is given by

$$M_{aero}(\alpha, \dot{\theta}) = qA_{ref}d_{ref}\left[C_{m_\alpha}\alpha + C_{m_q}\frac{d_{ref}}{2V}\dot{\theta}\right] \tag{19}$$

where $q = \frac{1}{2}\rho V^2$ is the dynamic pressure, A_{ref} and d_{ref} are the reference area and diameter, and V is the vehicle's velocity at the selected operating point. Linearizing about the nominal trajectory and including the canard control moment, the pitch equation of motion becomes

$$I_{yy}\ddot{\theta} + C_\theta\dot{\theta} + K_\theta\theta = K_\delta\,\delta \tag{20}$$

with

$$K_\theta = -qA_{ref}d_{ref}\,C_{m_\alpha} \tag{21}$$
$$C_\theta = -qA_{ref}d_{ref}^2\,C_{m_q}/2V \tag{22}$$
$$K_\delta = qA_{ref}d_{ref}\,C_{m_\delta} \tag{23}$$

Since the vehicle is statically stable, $C_{m_\alpha} < 0$, yielding $K_\theta > 0$ (restoring). Taking the Laplace transform of (20), the open-loop plant transfer function is

$$\frac{\Theta(s)}{\Delta(s)} = \frac{K_\delta}{I_{yy}s^2 + C_\theta s + K_\theta} \tag{24}$$

F. PID Controller Design

A PID control law was selected for the pitch loop:

$$\delta(t) = K_p\,e(t) + K_i\int e(t)\,dt + K_d\,\dot{e}(t), \qquad e = \theta_{ref} - \theta \tag{25}$$

Combining (24) with (25), the closed-loop transfer function is third-order due to the integral term:

$$\frac{\Theta(s)}{\Theta_{ref}(s)} = \frac{K_\delta(K_ds^2 + K_ps + K_i)}{I_{yy}s^3 + (C_\theta + K_\delta K_d)s^2 + (K_\theta + K_\delta K_p)s + K_\delta K_i} \tag{26}$$

To obtain a non-oscillatory step response, the natural extension of critical damping to a third-order system, all three closed-loop poles were placed at a common real location $-\sigma$, requiring the closed-loop characteristic polynomial to match $I_{yy}(s+\sigma)^3$. Equating coefficients with the denominator of (26) gives the controller gains as a function of the plant's own aerodynamic stiffness and damping:

$$K_d = \frac{3\sigma I_{yy} - C_\theta}{K_\delta} \tag{27}$$

$$K_p = \frac{3\sigma^2 I_{yy} - K_\theta}{K_\delta} \tag{28}$$

$$K_i = \frac{\sigma^3 I_{yy}}{K_\delta} \tag{29}$$

where σ is the single tuning parameter, selected to improve the settling-time while remaining within the deflection rate limit of the servomotor actuating the canards.

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Resulting Gain Values
Kp=0.00847   Ki=0.18267   Kd=0.0004
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However, because the aerodynamic coefficients vary with angle of attack and canard deflection, and because the available aerodynamic authority decreases with velocity, the final controller gains were obtained through direct numerical optimization rather than relying exclusively on the analytical pole-placement formulation. The optimization minimized the integrated squared pitch error while penalizing excursions in the opposite direction from the initial perturbation.

The resulting gains were then evaluated using the nonlinear aerodynamic representation obtained from CFD. In contrast with the linearized model used for controller design, the closed-loop simulation evaluates C_m directly from the interpolated aerodynamic data as a function of α and δ. This approach preserves the nonlinear variation of the aerodynamic moment within the range characterized by CFD.

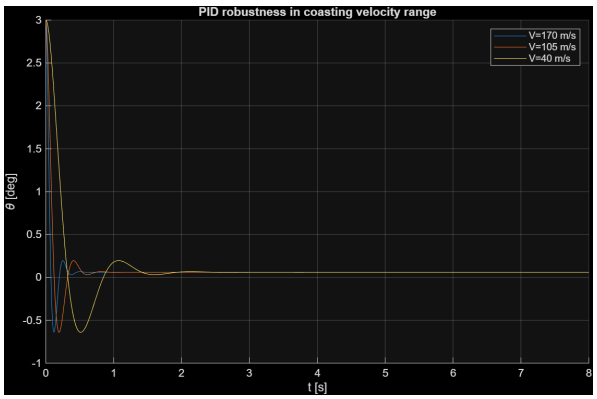


Fig. 11. PID Robustness with Velocity variations.

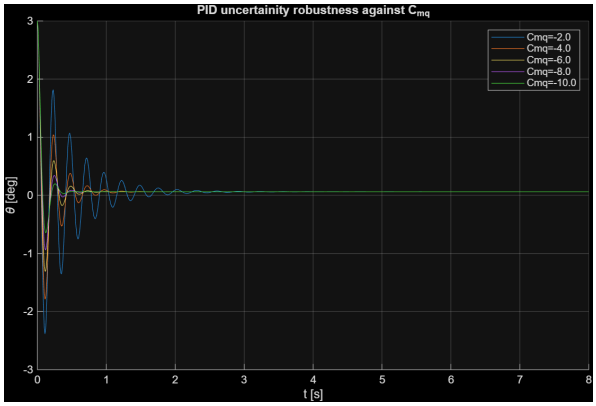


Fig. 12. PID Uncertainity with Different C_{mq} values.

Since the pitch-damping derivative C_{mq} was not directly obtained from the static CFD sweep and this would require unsteady or forced-oscillation simulations beyond the scope of this work a representative value was assumed for the closed-loop simulation, and its influence on controller performance was assessed through the robustness sweep shown in Fig 12.

G. Actuator Selection

The control-surface actuator was selected considering the constraints imposed by the limited internal volume of the vehicle, the modularity requirement of the avionics bay, the low-cost objective, and the need to provide repeatable angular positioning of the canards.

A TowerPro SG90-type micro servo was selected as the initial actuator for the prototype. The actuator is powered from a nominal 5 V supply and has a specified torque of approximately 1.8 kg·cm at 4.8 V.

The nominal torque can be converted to SI units using

$$1 \text{ kgf cm} = 0.0980665 \text{ N m}. \tag{30}$$

Therefore,

$$T_{servo} = 1.8(0.0980665) \approx 0.177 \text{ N m}. \tag{31}$$

The nominal angular response is specified as approximately 0.10 s per 60° at 4.8 V. The corresponding nominal angular speed is therefore

$$\omega_{servo} = \frac{60^\circ}{0.10 \text{ s}} = 600^\circ/\text{s}. \tag{32}$$

These specifications make the SG90 suitable as a low-cost prototype actuator from the standpoint of size, mass, power requirements, and nominal angular positioning capability.

However, the nominal servo torque cannot be directly compared with the vehicle pitching moment obtained from CFD. The latter represents the resultant aerodynamic moment about the vehicle reference point, whereas the servo must overcome the aerodynamic hinge moment acting about the canard rotation axis.

The actuator loading can therefore be represented generally as

$$T_{servo} \geq T_{hinge} + T_{friction} + T_{inertia}, \tag{33}$$

where T_{hinge} is the aerodynamic moment about the canard hinge, $T_{friction}$ represents mechanical friction and linkage losses, and $T_{inertia}$ represents the torque required to accelerate the control surface and linkage.

The hinge moment depends on the detailed geometry and location of the canard rotation axis. Consequently, the vehicle pitching moment M_y obtained from the CFD analysis cannot, by itself, be used as the required servo torque.

For this reason, the SG90 selection is considered a prototype-level actuator selection rather than a final demonstration of flight-worthiness. The final actuator validation requires experimental measurement of the control-surface loading and verification of the servo response under the expected mechanical loads.

H. Distinction Between Vehicle Pitching Moment and Actuator Torque

An important distinction must be made between the aerodynamic pitching moment of the complete vehicle and the torque required at the actuator.

The CFD quantity M_y represents the resultant moment acting on the vehicle about its reference axis. For example, the simulated configuration produced $M_y = 7.43$ and 12.26 for $\alpha = 0^\circ$ and canard deflections of 5° and 10°, respectively.

These values describe the aerodynamic authority of the control surfaces at the vehicle level. They do not represent the torque that must be supplied by the servo motor.

The actuator torque is instead determined by the aerodynamic moment about the canard hinge axis. In simplified form, this quantity can be expressed as

$$T_{hinge} \approx F_{aero} r_h, \tag{34}$$

where F_{aero} is the aerodynamic force acting on the control surface and r_h is the effective distance between the resultant aerodynamic force and the hinge axis.

Therefore, a complete actuator assessment requires knowledge of the hinge location and the pressure distribution over the canard. This distinction is particularly important when evaluating small servomotors, since a vehicle may have sufficient aerodynamic control authority while the selected actuator still lacks sufficient torque margin.

IV. HARDWARE IMPLEMENTATION

The avionics architecture was designed around low-cost, commercially available components, consistent with the requirement established in Section II. Attitude and altitude data are acquired by a magnetometer and a barometric pressure sensor, processed by a low-cost microcontroller, and logged onboard for post-flight analysis. Canard actuation is performed by standard hobby-grade servomotors, and a MOSFET-driven switching stage is used to control high-current loads independently of the microcontroller's logic-level outputs. The complete bill of materials is listed in Table III.

TABLE III

BILL OF MATERIALS FOR THE AVIONICS AND ACTUATION SYSTEM

Qty	Component	Function
4	SG90 servomotor	Canard actuation
2	18650 Li-ion cell	Power supply
1	Raspberry Pi Pico	Flight computer / control loop
1	MPU6050	IMU (gyroscope + accelerometer)
1	HMC5883	Magnetometer (heading reference)
1	BMP180	Barometric pressure / Temp. sensor
1	IRF520 MOSFET	Pyrotechnic recovery channel
1	SD card module	Onboard data logging
1	L7805 regulator	5 V power regulation

The Raspberry Pi Pico was selected as the flight computer due to its low cost, sufficient processing capability for a 100 Hz control loop, and native support for I2C and SPI peripherals, which simplifies interfacing with the sensor suite and the SD logging module. Attitude determination is performed by sensor fusion of the MPU6050 (angular rate and acceleration) and the HMC5883 (magnetic heading), providing both the pitch angle θ and its rate $\dot{\theta}$ used directly by the PID control law in Eq. (25). The BMP180 provides barometric altitude data for flight-phase detection and post-flight trajectory reconstruction. Power is supplied by two 18650 Li-ion cells, regulated to a stable 5 V rail via an L7805 linear regulator to power the microcontroller and sensor suite, while the IRF520 MOSFET stage independently switches higher-current loads such as the servomotor bank, isolating the actuation circuit from the microcontroller's logic-level supply.

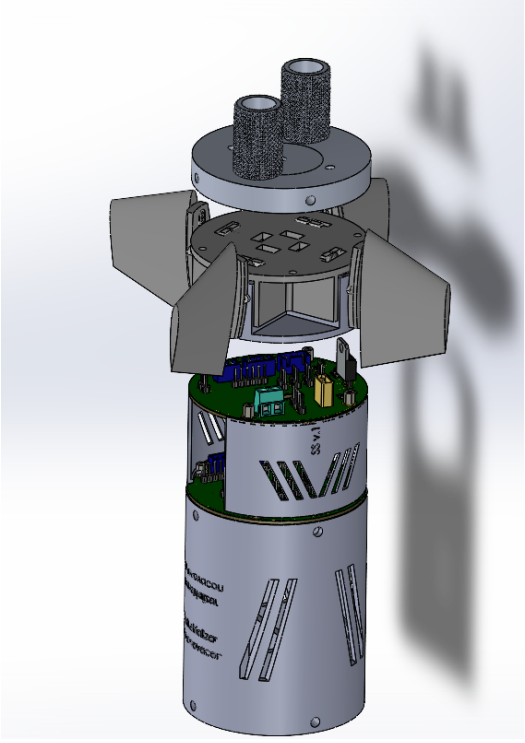


Fig. 13. Avionics Bay

V. Results and Discussion

The aerodynamic sweep obtained from CFD (Table I) shows a non-monotonic pitching-moment trend concentrated at $\alpha = 5°$, where C_{my} drops from 0.113 at $\delta = 0°$ to -0.007 at $\delta = 5°$ before rising sharply to 0.26 at $\delta = 10°$. A similar, less pronounced irregularity appears at $\alpha = -5°$. This behavior is reflected in the low coefficients of determination for these two sweeps in Table II ($R^2 = 0.302$ and 0.370, respectively), in contrast to the near-linear fits obtained at $\alpha \in \{-2, 0, 2\}$. Because the trend does not degrade monotonically with deflection, which would be the expected signature of progressive flow separation, this result is treated with caution: it may reflect genuine nonlinear aerodynamic coupling at high combined incidence, but it may equally reflect insufficient mesh resolution or non-convergence in these specific CFD cases, given the node/element limitations of the student ANSYS license noted in Section III-C. Resolving this ambiguity requires either a dedicated mesh-independence and convergence check for the $\alpha = \pm 5°$ cases, or wind-tunnel validation as proposed in Section VI, before this behavior can be confidently attributed to stall rather than numerical error.

A second effect observed across the simulated coast phase is the loss of control authority as vehicle velocity decreases. Since $K_\delta \propto \bar{q} = \frac{1}{2}\rho V^2$, the corrective moment available per degree of canard deflection drops considerably toward the end of the coasting window, requiring longer correction times for the same initial perturbation, as shown in the robustness sweep of Fig. 11. This is consistent with the physical expectation that active aerodynamic control degrades as dynamic pressure falls, and indicates that a constant-gain PID design is inherently more conservative at low velocity than at burnout conditions.

VI. Conclusions and Future Work

The system presented in this article meets its initial objective: an active pitch control prototype for a high-power rocket was designed, modeled, and validated in simulation using a systems engineering approach, from requirements definition through control synthesis and closed-loop verification. However, several phenomena identified during this process require further analysis before the system can be considered flight-ready.

The non-monotonic aerodynamic behavior observed at the combined $\alpha = 5°$, $\delta = 5°$ operating point, discussed in Section V, indicates that the deflection command should not be treated as a fixed actuator limit alone, but should instead be constrained as a function of the estimated angle of attack to avoid operating in the region of the aerodynamic envelope where this irregularity was observed. Similarly, the loss of control authority at low coast velocities suggests that a constant-gain PID may not be the most robust choice across the full range of the mission profile..

Based on these observations, the following improvements are proposed for future iterations of this work:

- **Actuator dynamics modeling.** The current plant model assumes an ideal, instantaneous actuator. Incorporating the SG90 servomotor's own response time and torque limitations into the simulation would provide a more realistic estimate of achievable settling time and would allow proper tuning via classical methods such as Ziegler–Nichols, which, as noted in Section IV, could not be meaningfully applied to the idealized plant.
- **Deflection scheduling near the stall boundary.** Rather than a fixed saturation limit, the maximum allowable canard deflection could be scheduled as a function of estimated angle of attack, keeping the vehicle within the linear aerodynamic region identified in the CFD sweep.
- **Gain scheduling with dynamic pressure.** Since control authority scales with $\bar{q}$, scheduling K_p, K_i, and K_d as functions of estimated velocity or altitude, rather than using constant gains, would maintain more consistent closed-loop performance throughout the coast phase.
- **Hardware-in-the-loop (HIL) testing.** The PID controller developed in this work is directly implementable on the Raspberry Pi Pico described in Section VI. The next validation step is software- and hardware-in-the-loop testing, exercising the real sensor suite and servomotors against the simulated plant before committing to a flight test.
- **Quaternion-based attitude estimation and LQR control.** A longer-term improvement path includes replacing the current Euler-angle attitude representation with a quaternion-based estimator (e.g., an Extended Kalman Filter fusing the MPU6050 and HMC5883 data) to avoid gimbal-lock and improve estimation accuracy, and evaluating a Linear-Quadratic Regulator (LQR) as an alternative to the PID architecture, particularly given the multi-input nature of a four-canard configuration.
- **Wind tunnel validation of the aerodynamic database.** Given that all control gains derive directly from the CFD-based C_m sweep, experimental validation of the stall boundary and control authority values against wind tunnel data would reduce the uncertainty currently carried into the control design. item **Flight testing.** As stated in Section I, this article is limited to ground-based validation; a flight test campaign remains the final step to validate the simulated closed-loop performance under real disturbances such as wind gusts and thrust misalignment.
- **Roll Control.** This paperwork assumed a perfect motion without roll moment, fact that is not real under flight conditions. This moment can create a different effect on trajectory due to canard deflection angle plane. This way, two of four canards can be used for roll moment while the rest are used for pitch control.

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